
UNIT 3 FUNCTIONS

Structure

- 3.0 Introduction
- 3.1 Objectives
- 3.2 Quantity
- 3.3 Interval
- 3.4 Function
 - 3.4.1 Mapping
 - 3.4.2 Pictorial Presentation of Functions
 - 3.4.3 Operations on Functions
 - 3.4.4 Some Commonly Used Functions
 - 3.4.5 Monotonic Functions
- 3.5 Types of Functions Based on Mapping
 - 3.5.1 Injective Function
 - 3.5.2 Surjective Function
 - 3.5.3 Bijective Function
 - 3.5.4 Geometrical Meaning of Injective, Surjective and Bijective Functions
- 3.6 Inverse of Functions
- 3.7 Summary
- 3.8 Answer to Check Your Progress
- 3.9 References and Further Readings

3.0 INTRODUCTION

Many times, we observe the association of the elements of one set with the elements of another set. For example, roll numbers of students in an examination of a university are associated with their corresponding marks. Such type of associations is discussed under the topic “Function”. In this unit, we will define function, explain its classification and various types. Function is a fundamental concept of Mathematics, viz. that of a **function**, which is a **special case of relation**, but **more general than that of 'operation'** on a set. While discussions in this unit, we will be using some concepts related to sets discussed in the unit -1 of this block.

3.1 OBJECTIVES

After completing this unit, you should be able to:

- define constant quantity, variable, interval, give some examples of each;
- define function, some particular functions;
- evaluate the value of some particular functions at given points;
- get an idea of one-one, onto and one to one correspondence and their geometrical interpretation; and

- finding inverse of a function.

3.2 QUANTITY

Before defining function, let us first explain what we mean by quantity, constant, variable and intervals.

Quantity

Here, by quantity we mean those things on which four basic mathematical operations addition, subtraction, multiplication and division can be applied. For example, temperature, height, weight, and time all these are quantities but they are continuous in nature, whereas books in a library, number of trees, number of balls is discrete in nature.

Note 1: If the nature of the quantity is such that it can take any possible value between two certain limits, then such a quantity is known as continuous in nature.

Let us consider the following example:

Suppose at the time of birth, height of a baby was 1.5 ft and after 15 years, the height of the same baby is 5 ft, then we know that height of this baby took all possible values between 1.5ft. to 5ft. That is, this is not the case that at the time of birth the height was 1.5 ft and in the next moment it reached to 1.6 ft and in successive moment to 1.7 ft. In fact, there are infinitely many values between 1.5 ft and 1.6 ft and all these values were taken by the height of that baby.

Note 2: Generally in mathematics and statistics, a variable is considered **discrete** if its possible values form a set of isolated points on a number line. If you can count the possible values—even if that counting goes on forever (countably infinite)—the variable is discrete.

Let us consider an example:

Number of children per family in a locality is an example of discrete quantity.

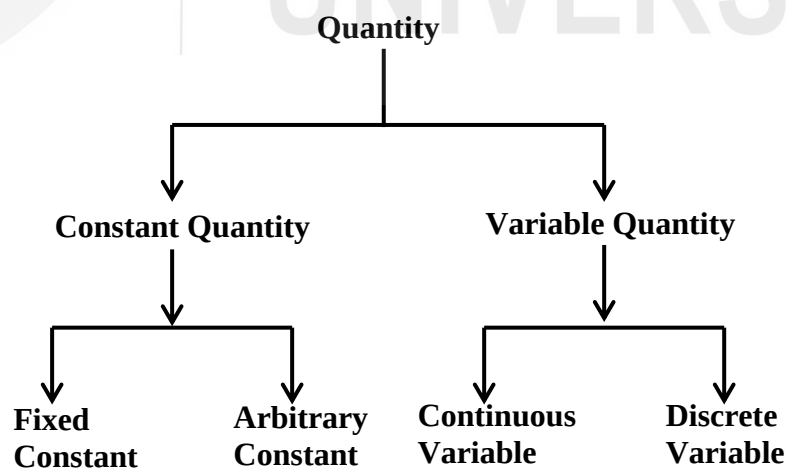


Fig. 3.1 : Representation Quantity

Constant Quantity

A quantity which remains same (unchanged) throughout a particular investigation is known as a constant quantity.

Fixed Constant

Those types of constants which always remain same (unchanged) independent of the purpose of user, place and time are known as fixed constants.

For example,

(i) $2, 5, \frac{3}{2}, \sqrt{13}, -\sqrt{17},$, etc.

(ii) quotient of circumference of a circle and its diameter which is always equal to π .

Arbitrary Constants

Those types of constants which remain same in one problem but may vary from problem to problem are known as arbitrary constants and are generally denoted by $a, b, c, l, m, n, \alpha, \beta, \gamma,$ etc.

For example, heights of the houses constructed by the different owners of the plots in a locality as per their own choice, is an example of arbitrary constant because heights of the houses vary from house to house as it depends on the choice of the owners.

Variable

A quantity which may change its value even in a particular problem is known as variable.

For example, blood pressures of a person as it varies time to time.

3.3 INTERVAL

Let R be the set of all real numbers. Then a set $I \subseteq R$ is said to be an interval if whenever $a, b \in I$ and $a < x < b$ then $x \in I$

For example, the set of all real numbers satisfying $2 \leq x \leq 3$ is an interval where x can take any real value between 2 and 3 including 2 and 3.

In unit 1 of this course, you have learned about Open Interval and Closed Interval, now let us learn some more intervals

Left Open and Right Closed Interval

A left open and right closed interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $(a, b]$ and is defined as

$$(a, b] = \{x \in R : a < x \leq b\}$$

In this case $x \in (a, b], \forall x, a < x \leq b$ and $a \notin (a, b],$ but $b \in (a, b]$

Left Closed and Right Open Interval

A left closed and right open interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $[a, b)$ and is defined as

$$[a, b) = \{x \in R : a \leq x < b\}$$

In this case $x \in [a, b), \forall x, a \leq x < b$ and $a \in [a, b),$ but $b \notin [a, b)$

Length of an Interval

Length of each of the intervals (a, b) , $[a, b]$, $(a, b]$, $[a, b)$ is defined as
 $b - a, a < b$

i.e. length of the interval = difference of the end points

For example, if $I = (2, 7)$ then $l(I) = 7 - 2 = 5$, where $l(I)$ denotes the length of the interval I .

Finite Interval

An interval is said to be finite if its length is finite.

For example, if $I = (-3, 5)$ then $l(I) = 5 - (-3) = 5 + 3 = 8$ which is finite.
 ∴ interval I is finite.

Infinite Interval

An interval is said to be infinite interval if its length is not finite.

For example,

(i) The set :

$$\{x \in R : x > a, a \in R\}$$

is an infinite interval and is denoted by (a, ∞)

(ii) The set $\{x \in R : x < a, a \in R\}$ is an infinite interval and is denoted by $(-\infty, a)$

Similarly, infinite intervals $[a, \infty)$, $(-\infty, a]$ are defined as

$[a, \infty) = \{x \in R : x \geq a\}$, where a is a fixed real number

$(-\infty, a] = \{x \in R : x \leq a\}$, where a is a fixed real number

Remarks 1:

- (i) Each interval contains infinitely many elements.
- (ii) Each interval is an infinite set but an infinite set may or may not be an interval. For example, N , W , Z , Q are infinite sets but are not intervals.
- (iii) A set may or may not be an interval.
- (iv) R , set of real numbers, is an infinite interval given by $R = (-\infty, \infty)$
- (v) Remember that $-\infty$ and ∞ are not included in the set of real numbers. Also, if extreme value is ∞ or $-\infty$ then open bracket is used on the side having extreme value.
- (vi) When we say that x is a finite number/real number it means that $-\infty < x < \infty$

Now we are in a position to define function.

3.4 FUNCTION

A function is a *mathematical* term, which is a tool used to describe some aspects of the real world, including the following:

a) Given that a vehicle moves at a constant average speed (say 50 km./hour), then the *distance* (say d) covered depends on the *time* (say t hours) over which the vehicle moves,

b) If a person takes a loan L (say, of Rs. 50,000/-) on simple interest R (say, at 6% per annum), the amount (say A) payable depends on the time T , after which the payment is made, *in addition to L and R .*

c) The area (say $Area$) of a circle depends on the radius (say r) of the circle.

The statements (a), (b) and (c) above express some *real* life situations, and each of which can be *mathematically* expressed respectively as (a) $d = 50 * t$, (a) $A = 50000 (1+6/100)^t$, and $Area = \pi.r^2$.

In each of these cases, there are two *variable quantities* involved: one on LHS and the other on RHS.

For example,

in (a) the quantity d occurs on LHS and quantity t occurs on RHS;

in (b) the quantity A occurs on LHS and quantity t occurs on RHS; and

in (c) the quantity $Area$ occurs on LHS and quantity r occurs on RHS.

For further discussion, first, we consider only (a) above. The value of d , occurring on L.H.S., is given by RHS, viz. $50*t$, and $50*t$ depends on the value of t *only*. For example, if $t = 3$ (hours), the distance covered, i. e. $d=50*t=50*3= 150$ km.; if $t= 5$ hours, then distance covered, i. e. $d = 50*5=250$ km.

Further, the dependence of $50*t$ on t is *unique* in the sense that for $t=3$, we get the *unique* value 150 or $50*3$, and for $t=5$, we get the *unique* value 250 or $50*5$. This unique *dependence* of $50*t$, and hence of variable quantity d is, in mathematical terms, called **functional** dependence, and $50*t$ is denoted as $f(t)$, for some f , and t inside shows that it depends on variable quantity t .

In view of the discussion above, f is called the **function name**, the variable quantity t is called the **independent variable** and the variable quantity d is called the **dependent variable**, *and the relation between d and t is called d 's functional dependence on t .*

Going back briefly to (b), $A = 50000 (1+6/100)^t$. Let $g(t) = 50000 (1+6/100)^t$, then $A = g(t)$. Here, g is a function name, t is independent variable, and A is the dependent variable, *and the relation between A and t is called A 's functional dependence on t .*

In case of (c), let $h(r) = \pi r^2$, then $Area = h(r)$, and h is a function name, r is the independent variable and $Area$ is the dependent variable, *and the relation between $Area$ and r is called $Area$'s functional dependence on r .*

Further, let X and Y be two sets. Then a rule which associates each element of X to a unique element of Y is called a function.

X is called domain of the function.

Y is called co-domain of the function and the set of only those values of Y for which function is defined is called range of the function.

That is, subset $\{y \in Y: y = f(x) \text{ for some } x \in X\}$ of Y is called range of the function.

Notation:

- (i) A function is generally denoted by f, g, h , etc., in the case of above definition we write $f: X \rightarrow Y$ and read as f is a function from X to Y .
- (ii) A function $f: X \rightarrow Y$ is generally described by writing $y = f(x), x \in X$, where $f(x)$ is an expression in terms of x .

A function is a special kind of relation. Functions are not restricted to sets of numbers only. For instance, let A be the set of mothers and B be the set of human beings. Then the rule that assigns to every mother her eldest child is a function. But the rule that assigns to **each mother her children** is **not a function** because it does not relate a unique element of B to each element of A .

In other words we can say that a function from a set, say A , to a set, say B , is a special type of relation where (i) each element $a \in A$ is in relationship with some element $b \in B$, and (ii) there is exactly one element $b \in B$ in relationship with each element $a \in A$.

More explicitly,

For two given *non-empty* sets X and Y , **a function f from X to Y** , denoted as

$f: X \rightarrow Y$ is a *rule* which associates to each element x of X a *unique* element, say, y of Y . The unique element y associated with x of X is denoted as $f(x)$, i. e. $y = f(x)$.

For the function $f: X \rightarrow Y$, the set X is called the domain of function f and Y is called the codomain of function f .

3.4.1 Mapping

If $f(x) = y$, then y is called image of x under f , and x is called *pre-image* of y . Also, the y , or $f(x)$, is called the *value* of x under the function f . *The names 'map' and 'mapping' are also used instead of the name 'function'.* The subset of Y which consists of all images $f(x)$ of elements of X is called the *range* of function f (in Y)

Remarks 2:

Important to note that the role/meaning of ' $\rightarrow$ ' is just a part of notation for a function. It just indicates that the symbol X , immediately on left of ' $\rightarrow$ ' is the *domain*, and the symbol Y , immediately on the right of ' $\rightarrow$ ' is the *codomain* of the function-symbol f , the symbol which precedes ':

Definitions: Let f be a function from A to B . The **set A is called the domain** of the function f and **B is called the co-domain** of function f . The set $\{f(x) | x \in A\}$ is called the range of f , and is also denoted by $f(A)$.

Given an element $x \in A$, the unique element of B to which the function f associates, it is denoted by $f(x)$ and is called the **f -image (or image) of x** or the value of the function f for x . We also say that f maps x to $f(x)$. The element x is referred to as the **pre-image of $f(x)$** .

For example, if $A = \{1,2,3,4\}$, $B = \{1,8,27,64,125\}$, and the rule f assigns to each member in A its cube, then f is a function from A to B . The domain of function f is A , its codomain is B and its range is $\{1,8,27,64\}$.

Can you tell what will be the domain and codomain for rule $f: f(x) = x/(1-x)$?

As you can see that $1-x=0$, if $x = 1$, in this case $f(x)$ will be undefined. Here domain of f can be taken as $\mathbb{R} \setminus \{1\}$ and codomain can be $\mathbb{R}$.

Definition: A function from a non-empty set A to a non-empty set B is a subset R of $A \times B$ such that for each $a \in A \exists$ a unique $b \in B$ such that $(a,b) \in R$. So, this relation satisfies the following two conditions:

for each $a \in A$, there is some $b \in B$ such that $(a,b) \in R$

if $(a,b) \in R$ and $(a,b') \in R$ then $b = b'$.

In previous examples we have seen that, we usually present functions as a rule associating elements of one set with another. So, let us present the definition again, with this view.

Definition: Let A and B be non-empty sets. A function (or a mapping) f from A to B is a rule that assigns to each element x in A exactly one element y in B . We write this as $f: A \rightarrow B$, read it as 'f is a function from A to B '.

Note that :

to each $a \in A$, f assigns an element of B ; and

to each $a \in A$, f assigns only one element of B .

So, for example, suppose $A = \{1,2,3\}$, $B = \{1,4,9,11\}$ and f assigns to each member in A its square values. Then f is a function from A to B . But if $A = \{1,2,3,4\}$, $B = \{1,4,9,10\}$ and f is the same rule, then **f is not a function** from A to B since no member of B is assigned to the element 4 in A .

Similarly, consider an example where $A = \{1,2,3\}$, $B = \{1,4,5,6,7\}$ and the rule f which associate $1 \rightarrow 1$, $2 \rightarrow 1$, $3 \rightarrow 4$. **Here f is not a function from A to B** since no member of B is assigned to the element 3 in A .

Note that the former example, $11 \in B$, but there is no element in A which is assigned to 11. **This does not matter**. It is not necessary that every element of B be related to some element of A .

Remarks 3: Each element of A has a unique image, and each element of B need not appear as the image of an element in A . Further, more than one element of A can have the same image in B .

Let us look at some examples of functions, and non-functions now.

There are two conditions for a rule to be a function.

- (i) Each element of X must be associated to some element of Y .
- (ii) There is unique element of Y corresponding to each element of X .

If b is a fixed element of B , then $f: A \rightarrow B : f(x) = b \forall x \in A$ is called a **constant function**.

Note that if $b=0$, then f is called the **zero map**, and is denoted by 0 .

$f: A \rightarrow A : f(x) = x \forall x \in A$ is called the **identity function**, and is denoted by **I**.

Also, in our discussion so far, in each case there is only ONE independent variable associated with each dependent variable. However, for a dependent variable, there may be **more than one independent variable**.

For example, the area of a rectangle of length l and breadth b is given by $\text{Area_Rectangle} = l*b$. The dependent variable Area_Rectangle depends on two independent variables, viz. l and b . It may be written as $\text{Area_Rectangle} = f_1(l, b)$, where $f_1(l, b) = l*b$. Note the comma in $\text{Area_Rectangle} = f_1(l, b)$.

Similarly, the volume v of a railway container of length l , breadth b and height h is given by $v = l*b*h$. Then the volume v may be written as a function $f_2(l, b, h)$ of three independent variables l, b , and h .

Let us look at some more examples for illustrating the concept of function.

i) Let $X = Y = N$, the set of natural numbers, then

$f: N \rightarrow N$, such that $f(x) = 2x+1, \forall x \in N$.

Clearly, for each $x \in N$, $2x+1$ exists in N and is unique for each x .

Hence, $f: N \rightarrow N$ is a function.

ii) Let $f: Z \rightarrow N$, such that $f(x) = x^2+3$,

then for each $x \in Z$, $x^2+3 \in N$ and is unique, therefore, f is a function

iii) Let $f: Z \rightarrow N$, such that $f(x) = x^2-3$,

then for $x=0$, $x^2-3=-3 \notin N$, Hence, $f: Z \rightarrow N$ is **not a function** from Z to N .

iv) Let $f: Z \rightarrow Z$, such that $f(x) = x^2-3 \in Z$

then $x^2-3 \in Z$ (codomain), $\forall x \in Z$ (domain). Hence, $f: Z \rightarrow Z$ is a function from Z to Z .

Let us see some examples:

Example:

Function Name: It is indeed called the **greatest integer function** or, more commonly in modern mathematics, the **floor function**.

Domain and Codomain: The function correctly maps real numbers (R) to integers (Z).

We denoted this function's action by $f(x)=[x]$, where $[x]$ is the greatest integer $\leq x$.

Some more examples:

- For $x=0.6$, the greatest integer less than or equal to 0.6 is **0**.
- For $x=2.3$, the greatest integer less than or equal to 2.3 is **2**.
- For $x=-5$, the greatest integer less than or equal to -5 is **-5**.

Example:

Function $f: \mathbb{R} \rightarrow \mathbb{R}: f(x) = |x|$ is known as the modulus (or absolute value) function, where $|x|$ is the absolute value of x .

For example, if $x=10$ then $f(x) = |x| = 10$ and if $x = -10$, then $f(x) = |x| = 10$.

Now let us take sets, $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4, 5\}$. Consider the rule f which associates $a \rightarrow 1, a \rightarrow 3, b \rightarrow 2, c \rightarrow 3$. This is **not a function** from A to B because, elements 1 and 3 $\in B$ are assigned to the same element $a \in A$.

3.4.2 Pictorial Presentation of Functions

A function can be presented diagrammatically. Here we will see few examples of that. In next unit of this block, you will learn in more detail about diagrammatical representation of different types of functions.

Consider $A = \{1, 2, 3, 4\}$, $B = \{1, 4, 5\}$ and the rule f which associates $1 \rightarrow 1, 2 \rightarrow 4, 3 \rightarrow 5, 4 \rightarrow 5$. **Then f is a function from A to B .**

Pictorial representation of above function f :

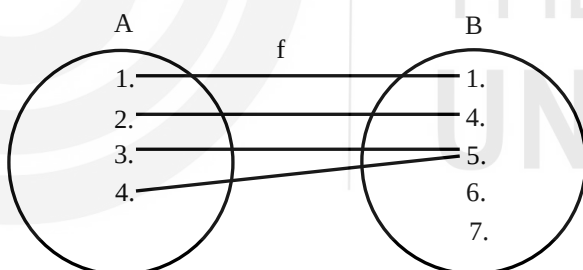


Fig. 3.2: The rule f is a function

Let us take another example of function f where $f(x) = x+5$. If we take domain for this function $\{1, 2, 3, 4\}$ then diagrammatic representation will be :

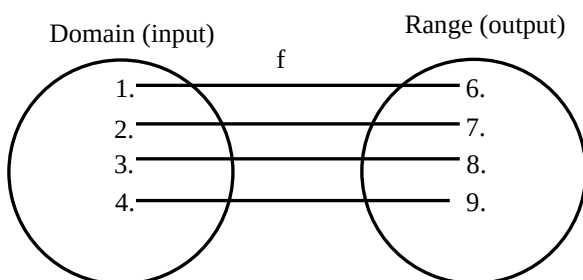


Fig. 3.3: Diagrammatic representation function $f(x) = x+5$

Check Your Progress-1

1). Let $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = 3n, n \in \mathbb{N}$

Express the function diagrammatically. Also, write domain, range and co-domain of the function.

.....

2). If $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$f(x) = 3^x, x \in \mathbb{R}$, then obtain (i) Domain of f (ii) Range of f

.....

3). Find the domain and range of the following real-valued real functions:

(i) $f(x) = -|x|$ (ii) $f(x) = +\sqrt{9-x^2}$

.....

4). For each of the following, tell whether f is a function or not. Justify, your answer.

i) $f: \mathbb{N} \rightarrow \mathbb{N}$, such that $f(x) = x - 3$

ii) $f: \mathbb{N} \rightarrow \mathbb{Z}$, such that $f(x) = x - 3$

.....

5). Find the domain of the function, $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$f(x) = \sqrt{(x-3)(5-x)}, x \in \mathbb{R}$

Also evaluate $f(3), f(4), f(5)$.

.....

6). Find the domain and range of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) =$

$\frac{1}{x+3}, x \in \mathbb{R}$.

Also, evaluate $f(0), f(2), f(-3)$ if possible.

.....

7). Every function is a relation. Is every relation a function? Why?

.....

8). Consider the following pseudocode:

```

read(n)
while n > 1 do
begin
if n is even then n := n div 2
else n := 2n + 1;
end

```

Write a function of n that describes the operations performed.

.....

Some more Concepts Related to Function are given as under

- (i) If $f: X \rightarrow Y$ is a function given by $y = f(x)$ then x is known as a pre image of y and y is known as image of x .
- (ii) If $y = f(x)$ is a function then values of y depend on values of x . So, y is known as dependent variable and x is known as independent variable.

Let us now take up some examples which will enable you to distinguish as to whether a rule is a function or not.

For example,

(a)

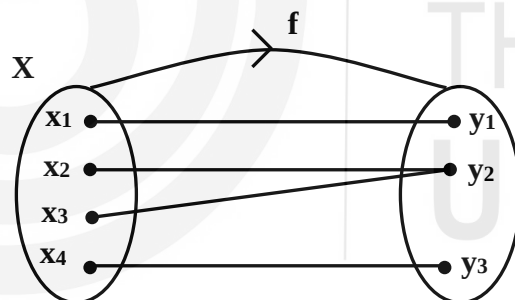


Fig. 3.4: A function

The rule f shown in Fig. 3.4 is a function because it satisfies both conditions i.e.

- (i) Each element of X is associated to some element of Y
- (ii) There is unique element of Y corresponding to each element of X .
 i.e. y_1 is unique element of Y corresponding to $x_1 \in X$
 y_2 is unique element of Y corresponding to both $x_2, x_3 \in X$
 y_3 is unique element of Y corresponding to $x_4 \in X$

Here domain of $f = \{x_1, x_2, x_3, x_4\}$

Co-domain of $f = \{y_1, y_2, y_3\}$ and range of $f = \{y_1, y_2, y_3\}$

(b)

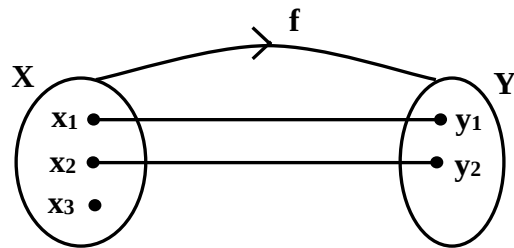


Fig. 3.5: Not a function

The rule f shown in Fig. 3.5 is not a function because $x_3 \in X$, but it is not associated to the element of Y .

Out of two restrictions for a rule to be a function, first is that each element of X must be associated to some element of Y .

(c)

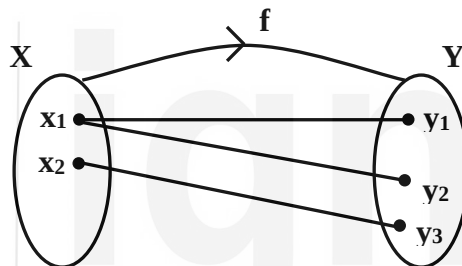


Fig. 3.6: Not a function

The rule f shown in Fig. 3.6 is also not a function because $x_1 \in X$ is not associated to unique element of Y .

For a rule to be a function it is must that each element of X is associated to a unique element of Y .

(d)

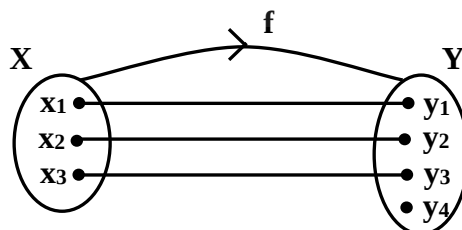


Fig. 3.7: A function

The rule f shown in Fig. 3.7 is a function because it satisfies both the conditions for a rule to be a function, i.e.

- (i) Each element of X is associated to some element of Y .
- (ii) There is unique element of Y corresponding to each element of X .

Here domain of the function $f = \{x_1, x_2, x_3\} = X$

Co-domain of the function $f = \{y_1, y_2, y_3, y_4\} = Y$

Range of the function $f = \{y_1, y_2, y_3\} \subset Y$

Real valued function

A **real-valued function** is a function where the **output (the "value")** is always a real number.

In mathematical notation, if we have a function $f: X \rightarrow Y$, it is a real-valued function if the codomain (Y) is the set of real numbers ($\mathbb{R}$) or a subset of it.

Remark :

1. **Real-Valued Function:** Only the **output** must be a real number. The input (domain) could be anything—points on a map, people, or complex numbers.
2. **Real Function:** Both the **input** and the **output** must be real numbers.

Now let us see how different operations like addition, subtraction, multiplication and division can not only be applied only on real valued functions.

Here it is important to note that the "Algebra of Functions" is commonly taught using real-valued functions, these operations can be applied to many other types of functions as well.

The ability to add, subtract, multiply, or divide functions depends on whether the **outputs** of those functions belong to a mathematical structure that allows those **operations (like a Field, Ring, or Vector Space)**.

For real-valued functions f and g , the operations are defined **pointwise**. This means we perform the math on the outputs for every shared input x .

3.4.3 Operations on Functions

The *binary operation* '+' on Natural numbers may be considered for the function **Plus**: $N \times N \rightarrow N$, such that

Plus (x, y) = ($x+y$), for each pair of Natural numbers.

The union operation $X \cup Y$ on sets can be considered as a function

Union: $P \times P \rightarrow P$, where P is a set of all subsets of universal set U , such that **Union** (X, Y) = $X \cup Y$, for each pair X, Y of sets from P .

In general, an operation is a function in which the domain of function is either the codomain or cross-product of codomain of function, Cross-product may involve the codomain finitely many times: Codomain $\times$ Codomain ... $\times$ Codomain.

For example, the square operation on N is the function $SQ: N \rightarrow N$, such that $SQ(n) = n^2$. The domain of function SQ is the codomain of the function SQ . In the case of Plus and Union, just shown above, the Domain is a cross- product of codomain.

If given whose domain ranges are subsets of the real numbers, we define the function $f+g$ by $(f + g)(x)$ to be the function whose value at x is the sum of $f(x)$ and $g(x)$. Symbolically,

$(f + g)(x) = f(x) + g(x)$. This is called pointwise addition of f and g .

The domain of $f+g$ is the intersection of the domains of f and g since to compute $(f + g)(x)$ it is necessary and sufficient to compute both $f(x)$ and $g(x)$.

Other operations on functions are defined similarly:

- $(fg)(x) = f(x)g(x)$ (pointwise multiplication)
- $f^p(x) = (f(x))^p$ for any real exponent p with the domain of f^p consisting of those points for which the p -th power of $f(x)$ makes sense.
- $(f/g)(x) = f(x)/g(x)$, for $g(x) \neq 0$ (pointwise multiplication)

For example, if $f(x) = 3 \sin(x)$ and $g(x) = x^2$, then

$$(f+g)(x) = 3 \sin(x) + x^2$$

$$(fg)(x) = 3 \sin(x) * x^2$$

$$(f-g)(x) = 3 \sin(x) - x^2$$

$$(f/g)(x) = 3 \sin(x) / x^2$$

The domains of both f and g are all real numbers, but the domain of f/g is $\{x \mid x \neq 0\}$.

Now let us consider two functions f and g from $A = \{1,2\}$ to, $B = \{1,2,3,4\}$, where $f=\{1,1\},(2,4)\}$. Let g be defined by the rule $g(x)=x^2$ where the domain of g is the set $\{1,2\}$. Here both have the same domain. Since f and g assign the same image to each element in the domain, they have the same effect throughout. This is why we treat them as the same, or equal.

Definition: If f and g are two functions defined on the same domain A and if $f(a)=g(a)$ for every $a \in A$, then the functions f and g are equal, i.e., $f = g$.

Let us take another example $f(x) = x^2 + 5$, where x is a real number, and $g(x) = x^2 + 5$, where x is a complex number. Then the function f is not equal to the function g since they have different domains although $f(x)=x^2 + 5= g(x) \forall x \in \mathbb{R}$. By this example we can conclude that even if $f(a)= g(a)$, f and g may not be the same.

So far, the operations you have seen are the same as those for member systems.

However, there is yet another operation on functions which we now define.

Definition: Let f and g be the operation of combining two functions by applying them one after the other. That is, the composition of $f(x)$ and $g(x)$, denoted by, $f \circ g$.

For example, consider $f : \mathbb{R} \rightarrow \mathbb{R}$: $f(x) = (x^3 + 2x)^3$. We can write it as the composition of g and h , where the value of $f(x)$ can be obtained by first calculating $x^3 + 2x$ and then taking its third power. We can write g for first or inside function $g(x) = x^3 + 2x$. We write h for the second function: $h(x) = x^3$. The use of the variable x is irrelevant, we could as well write $h(y) = y^3$ for $y \in \mathbb{R}$. We can see that $g \circ h(x) = g(x^3) = (x^3 + 2x)^3 = f(x)$.

In general $(f \circ g) \neq (g \circ f)$.

For example, if $f(x) = x^2$ and $g(x) = x+1$, then $(f \circ g)(x) = (x+1)^2$ and

$$(g \circ f)(x) = x^2 + 1.$$

Here we can see that $f \circ g \neq g \circ f$.

Function Composition is Not Commutative. This property means that the **order** in which you compose functions matters.

Function Composition is Associative:

Let us see another example, where $f(x) = x^2$, $g(x) = x+1$, $h(x) = x^3$

Here, $(f \circ (g \circ h))(x) = (x^3 + 1)^2$ and $((f \circ g) \circ h)(x) = (x^3 + 1)^2$. Here we can see $f \circ (g \circ h) = (f \circ g) \circ h$.

This property means that when you are composing three or more functions, the **grouping** of the functions does not matter.

Let us break down the calculation for $f(x)=x^2$, $g(x)=x+1$, and $h(x)=x^3$:

1. Calculating $f \circ (g \circ h)$

- First, we find the inner function $(g \circ h)(x)$: $(g \circ h)(x) = g(h(x)) = g(x^3) = x^3 + 1$
- Now, we compose f with that result: $(f \circ (g \circ h))(x) = f(x^3 + 1) = (x^3 + 1)^2$

2. Calculating $(f \circ g) \circ h$

- First, we find the inner function $(f \circ g)(x)$:
 $(f \circ g)(x) = f(g(x)) = f(x+1) = (x+1)^2$
- Now, we compose that result with $h(x)$ (by substituting $h(x)$ for x):
 $((f \circ g) \circ h)(x) = (h(x) + 1)^2 = (x^3 + 1)^2$

Both methods produce the exact same result, confirming that the composition is **associative**.

3.4.4 Some Commonly Used Functions

Based on the nature of classification a function may be given some particular names. In this section you will see some commonly used function names, their definitions and their graphical representations. More on graphical representations of different types of functions you will learn in next unit of this Block.

Constant Function

Let $X, Y \subseteq \mathbb{R}$, then a function $f: X \rightarrow Y$ is said to be a constant function if it is defined as: $f(x) = a, \forall x \in X$, where a is a real constant.

i.e. a function is constant if range is a singleton set.

i.e. all elements of the domain are associated to a single element of the co-domain of the function.

For example,

(i) $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 3, x \in \mathbb{N}$

is a constant function because all elements of the domain are associated to the single element 3 as shown in the Fig. 3.8

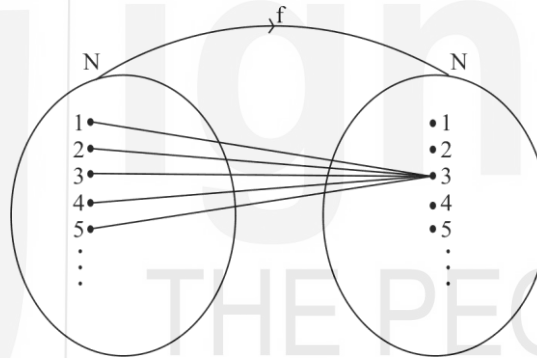


Fig. 3.8: Constant Function

(ii) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2, x \in \mathbb{R}$ is also a constant function and its graph is given below in Fig. 3.9.

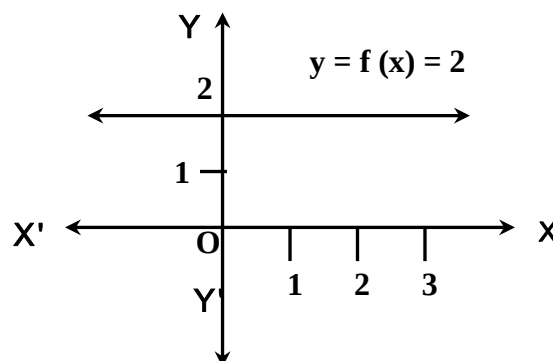


Fig. 3.9: Constant Function

Identity Function

Let $X \subseteq \mathbb{R}$, a function $f: X \rightarrow X$ is said to be an identity function if it is defined as :

$$f(x) = x, \quad x \in X$$

i.e. a function is said to be identity function if each element is associated to itself.

For example,

- (i) $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n, \quad n \in \mathbb{N}$ is an identity function as shown in the Fig. 3.10.

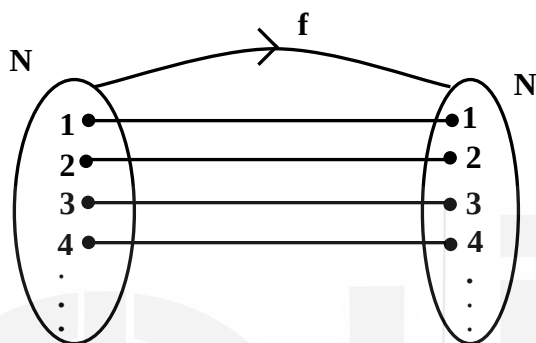


Fig. 3.10: Identity Function

- (ii) Function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x, \quad x \in \mathbb{R}$ is also an identity function and its graph is given in Fig. 3.11.

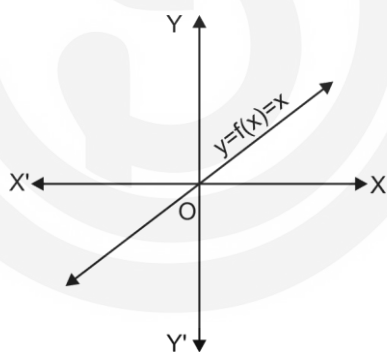


Fig. 3.11: Identity Function

3.4.5 Monotonic Functions

When a function that is either entirely non-increasing or non-decreasing it is called monotonic function.

For Example,

Increasing: If $x \leq y$, then $f(x) \leq f(y)$.

Any straight line with a positive slope is an increasing function. The general form is $f(x) = mx + c$, where the slope m is greater than 0.

Strictly Increasing: If $x < y$, then $f(x) < f(y)$.

Note that strictly increasing functions are always injective.

It is commonly found that some functions to be increasing only on specific portions of their domain.

For Example $f(x) = x^2$: This parabola is increasing for all $x > 0$. (It is decreasing for $x < 0$).

Linear Function : Any line with a positive slope, like $f(x) = 2x+1$. For every step you take to the right on the x-axis, you go up on the y-axis.

Exponential Function: The natural exponential function $f(x) = e^x$. This function not only increases but does so at an ever-faster rate.

Logarithmic Function: The natural logarithm $f(x) = \ln(x)$, $x > 0$. This function increases for all positive x values, though its rate of increase slows down as x gets larger.

Check Your Progress -2

1. Let $f(x) = 1/x$ and $g(x) = x^3 + 2$. Find the following functions, where $x \in \mathbb{R}$.

$$(f + g)(x)$$

$$(f - g)(x)$$

$$(fg)(x)$$

$$(f/g)(x)$$

.....

2. Let $f(x) = \sqrt{x+1} \quad \forall x \geq -1$ and $g(x) = x^3 \quad \forall x \in \mathbb{R}$. Define the following functions. Also give their domains.

$$(f + g)(x)$$

$$(f - g)(x)$$

$$(fg)(x)$$

$$(f/g)(x)$$

$$(f \circ g)(x)$$

.....

3. If $f(x) = x^2$, $g(x) = x+2$ and $h(x) = x^3 + x+2$ find $g \circ f$ and $f \circ g \circ h$

.....

3.5 TYPES OF FUNCTIONS BASED ON MAPPING

Now let us discuss some types of functions based on how inputs are mapped to outputs. In other words it describes how the inputs in the domain relate to the outputs in the codomain.

By definition of function $f: X \rightarrow Y$, for each x , its image $f(x)$ must belong to Y and $f(x)$ should not have more than one value in Y , as happens in the case of $f(x) = \text{square-root of } x$. **However, there is no restriction on elements of X , for f to be a function.** For example, for the function Mod , and for both values 5 & -5 , $\text{Mod}(5) = \text{Mod}(-5) = 5$. So mapping of function helps in deciding its type. Let us look at different types of mappings.

3.5.1 Injective Function (One-One Function)

A function $f: X \rightarrow Y$ is called injective, if for x_1, x_2 in X with $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

The definition can be equivalently stated as:

A function $f: X \rightarrow Y$ is called injective, if $f(x_1) = f(x_2)$ then $x_1 = x_2$

Note: The term 1-1 is also used instead of **injective**. However, there is a similar term 1-to-1 used for a distinct concept viz. **bijective**. **For this reason, the terms 1-1 and 1-to-1 may be avoided.**

Some examples:

i) The function $f: \mathbb{N} \rightarrow \mathbb{N}$, such that $f(x) = 2x+1, \forall x \in \mathbb{N}$ is injective. Let $f(x) = f(y)$, then $2x+1 = 2y+1$ implying $x = y$

ii) The function $f: \mathbb{N} \rightarrow \mathbb{R}$, defined by the rule $f(x) = \text{positive square-root of } x$ is injective. Let $f(x) = f(y)$, i. e. $\sqrt{x} = \sqrt{y}$ squaring both sides, we get $x = y$.

iii) The function $f: \mathbb{N} \rightarrow \mathbb{R}$, defined by the rule $f(x) = x^2$ is injective, as, for x, y from $\mathbb{N}$, if $f(x) = f(y)$, i. e. $x^2 = y^2$ then $x = y$ or $x = -y$, but as y is in $\mathbb{N}$, the number $-y$ cannot be in $\mathbb{N}$. Hence, $x = y$.

Procedure (generally adopted) to establish a function as injective:

For function $f: X \rightarrow Y$,

i) assume that $f(x) = f(y)$ and then show $x = y$ (must) for $x, y \in X$ or alternatively, in order to establish **f is NOT injective**, show for **some** pair of elements x, y , with $x \neq y$ of X , that $f(x) = f(y)$.

The latter method was used in showing $f: \mathbb{Z} \rightarrow \mathbb{R}$, defined by the rule $f(x) = x^2$, is not injective by using $f(2) = f(-2) = 4$. **We have already used the first one to show** the function $f: \mathbb{N} \rightarrow \mathbb{R}$, defined by the rule $f(x) = x^2$ as injective.

Functions

Injective Mapping: A mapping $f: A \rightarrow B$ is said to be injective (or one-one) if the images of distinct elements of A under f are distinct, i.e., if $x_1 \neq x_2$ in A , then $f(x_1) \neq f(x_2)$. So, f is 1-1.

In other words a function $f: X \rightarrow Y$ is said to be 1-1 or **injective** function if distinct elements of X are associated to distinct elements of Y under f .

i.e. if $x_1, x_2 \in X$ be s.t.

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Or if $x_1, x_2 \in X$ and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

If we compare this definition with example of daughters and mothers then one-one function means each daughter must have different mother, i.e. there cannot be two daughters having same mother for a function to be one-one.

For example,

(i)

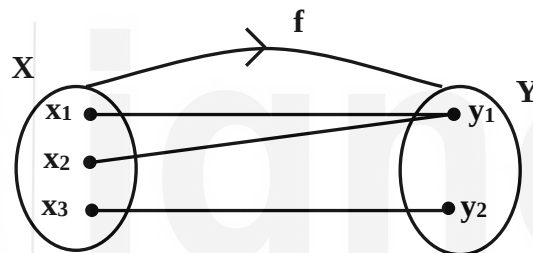


Fig. 3.12: Not a 1-1 function

The function f shown in Fig. 3.12 is **not one-one** functions because two different elements x_1, x_2 of X have the same image y_1 .

(ii)

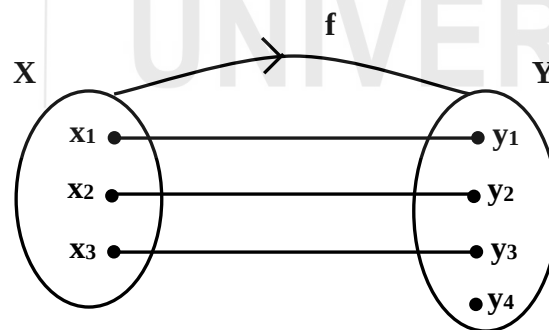


Fig. 3.13: A 1-1 function

The function f shown in Fig. 3.13 is one-one function because all the three elements of X have distinct images in Y .

Example:

Show that the function $f: R \rightarrow R$ defined by $f(x) = 7x + 5$ is 1-1 function.

Solution: Let x_1, x_2 be s.t.

$$f(x_1) = f(x_2) \Rightarrow 7x_1 + 5 = 7x_2 + 5 \Rightarrow 7x_1 = 7x_2 \Rightarrow x_1 = x_2$$

$\Rightarrow f$ is 1-1 function

3.5.2 Surjective (Onto) Function

For a function $f: X \rightarrow Y$, it is essential that every element of X (**domain**) corresponds to some element of Y . But, for f to be a function, it is not compulsory that for every y of Y (**codomain**), there be a corresponding element x of X such that $f(x) = y$.

Onto Mapping: A mapping $f: A \rightarrow B$ is said to an onto (or surjective) mapping if $f(A) = B$, that is, the range and co-domain coincide. In this case we say that f maps A onto B .

For example, $f: Z \rightarrow Z : f(x) = x+1, x \in Z$, then every element y in the co-domain Z has a pre-image $y-1$ in the domain Z . Therefore, $f(Z) = Z$, and f is an onto mapping.

If we take another example, for the function $f: N \rightarrow N$, with $f(x) = 2x+1 \quad \forall x \in N$, for the element 4 of N (the codomain) an x in N (the domain), if exists, satisfies $2x+1 = 4$, i. e. $x = 3/2$, which is not in N . **Hence f is not surjective.**

A function $f: X \rightarrow Y$ is called **surjective**, if for an arbitrary element y of Y (**the codomain**), we are able to find an x in X (**the domain**) such that $f(x) = y$.

Procedure for establishing a function as surjective/onto:

Take an arbitrary element y of the **codomain** Y , and show the existence of some element x of domain X with $f(x) = y$. If it is possible to find such an element x of **domain** X , then the function is surjective, *else*, it is not surjective.

Example: For the function $\text{Mod}: Z \rightarrow N$ such that:

$$\text{Mod}(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Let x be in the codomain N , then this x belongs to Z , the domain (*because $N \subseteq Z$*) with $|x|=x$, hence, Mod is surjective.

Next, consider $f: Z \rightarrow C$, defined by the rule $f(x) = \text{positive square-root of } x$.

Let, for x in Z , with $f(x) = 3.75$, i. e. positive-square-root of $x = 3.75$. Therefore, $x = (3.75)^2$, which is not in Z . Hence, **$f: Z \rightarrow C$ is not surjective**

A function $f: X \rightarrow Y$ is said to be onto or **surjective** if each element of Y has at least one pre image in X .

i.e. for each $y \in Y$, there exists at least one $x \in X$ such that $f(x) = y$

If we compare this definition with example of daughters and mothers then onto function means each mother must have at least one daughter.

For example,

(i)

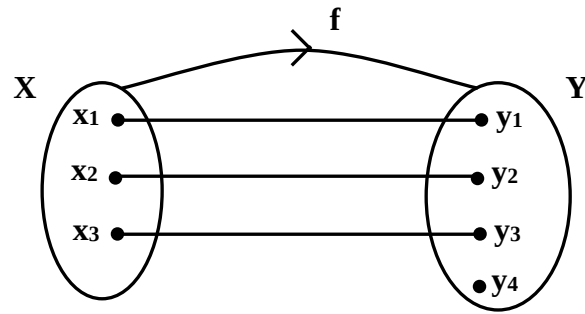


Fig. 3.14: Not an onto function

The function f shown in Fig. 3.14 is not onto function because $y_4 \in Y$ but there is no $x \in X$ such that: $y_4 = f(x)$

(ii)

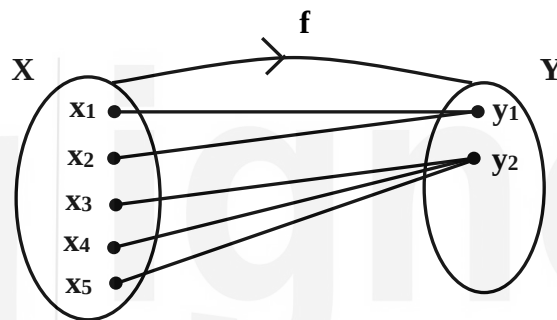


Fig. 3.15: An onto function

The function f shown in Fig. 3.15 is onto function because each element of Y has at least one pre image, i.e. y_1 has two pre images and y_2 has three pre images.

Remark 4: If we want to show that a function $f(x)$ is onto then first we take an element y in Y and we have to show that there exists an element x in X such that $f(x) = y$

For example, show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 7x + 5$ is onto function.

Solution: Here $X = \mathbb{R}$, $Y = \mathbb{R}$

Let $y \in Y = \mathbb{R}$, then $\frac{y-5}{7} \in X = \mathbb{R}$ s.t.

$$f\left(\frac{y-5}{7}\right) = 7\left(\frac{y-5}{7}\right) + 5 = (y-5) + 5 = y$$

$\therefore f$ is an onto function.

3.5.3 Bijective (One-One and Onto) Function

A function $f: X \rightarrow Y$ is called Bijective if and only if it is both injective and surjective. A bijective function from X to Y is also called **1-to-1 correspondence** between X and Y .

Bijjective Mapping: A Member $f: A \rightarrow B$ is said to be bijective (or one-one onto,) if f is both injective and surjective, i.e., one-one as well as onto.

For example, $f: \mathbb{Z} \rightarrow \mathbb{Z} : f(x) = x+2, x \in \mathbb{Z}$ is both injective and surjective. So, f is bijective.

Let us take another example: $f(x)=x-4$.

This function is one-to-one (different inputs give different outputs) and onto (any real number output y can be achieved with the input $x=y+4$), so, f is bijective.

A function $f: X \rightarrow Y$ is said to be one-one and onto or **bijective** or **one-one correspondence** if:

- (i) f is one-one
- (ii) f is onto

If we compare this definition with example of daughters and mothers then one-one and onto function means each mother have exactly one daughter and there is no mother who does not have any daughter. The function f shown in Fig.3.16 represents a situation of one-one and onto function.

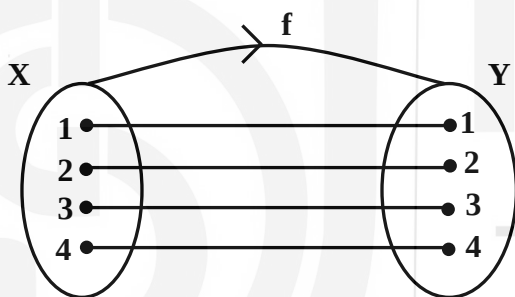


Fig. 3.16: one – one and onto function

3.5.4 Geometrical Meaning of Injective, Surjective and Bijjective Functions

One-One Function

If a function is 1-1 then geometrically it will satisfy the following condition. Each horizontal line either does not intersect the graph of the function or if it intersects it will intersect exactly at one point.

For example,

- (i) Graph shown in Fig. 3.17 (b) is the graph of a one-one function because each horizontal line either does not intersect the graph or if it intersects, it will intersect exactly at one point, i.e. each horizontal line above x-axis will intersect the graph exactly at one point and each horizontal line below x- axis not intersect the graph at all.
- (ii) Graph shown in the Fig. 3.17 (c) is also the graph of a one-one function because each horizontal line intersects the graph exactly at one point.
- (i) But the graph shown in the Fig. 3.17 (a) is not the graph of a one-one function because if we draw any horizontal line below the x-axis, then it will intersect the graph at two points. Similarly graph shown in Fig. 3.17 (d) is also not a one-one function.

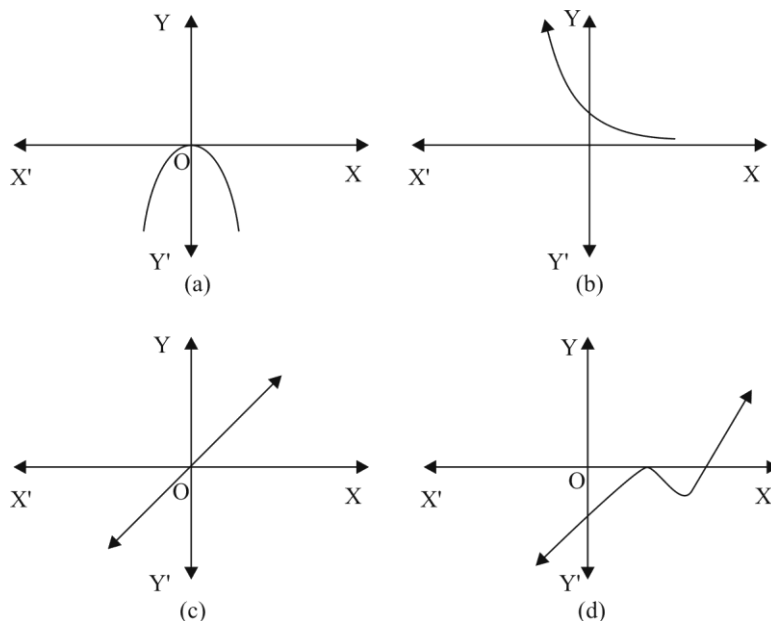


Fig. 3.17: Some functions

• Onto or Surjective Function

If a function is onto then geometrically it will satisfy the following condition. Each horizontal line must intersect the graph of the function at least at one point.

For example,

- (i) Graph shown in the Fig. 3.17 (d) is the graph of an onto function because each horizontal line intersect the graph of the function at least at one point. Graph shown in Fig. 3.17 (c) is also onto function because each horizontal line intersects the graph exactly at one point.
- (ii) But the graph shown in the Fig. 3.17 (b) is not the graph of a surjective function because if we draw any horizontal line below the x-axis, then it will not intersect the graph.
- (iii) Similarly, the graph shown in the Fig. 3.17 (a) is not the graph of an onto function because if we draw any horizontal line above the x-axis, then it will not intersect the graph.

• Bijective Function

If a function is bijective or one-one and onto or one-one correspondence, then each horizontal line must intersect the graph of the function exactly at one point.

For example,

- (i) Graph shown in the Fig. 3.17 (c) is the graph of a bijective function because each horizontal line intersects the graph of the function exactly at one point.
- (ii) But the graphs shown in the Fig. 3.17 (a) and (b) are not the graphs of a bijective function because they are not onto.

- (iii) Similarly, the graph shown in the Fig. 3.17 (d) is not the graph of a bijective function because it is not one-one.

3.6 INVERSE OF FUNCTIONS

Purpose of an **inverse of a function** is to "**reverses**" or "**undo**" the action of another function. For example if you take an input, apply a rule/function ($f(x)$) to it, and then apply the inverse function ($f^{-1}(x)$) to the output, you get your original input back. Here function $f(x)$, its inverse is denoted as $f^{-1}(x)$. Here it is very important to note that the "-1" is not an exponent but it is just notation for the inverse.

The key idea while using inverse function is that if $f(a) = b$, then $f^{-1}(b) = a$.

Example:

Let us say the function $f(x)=3x+5$ takes an input, triple it, and adds 5.

Then for $x = 2$; $f(2)=3(2)+5=11$

Then no need to say that inverse function, $f^{-1}(x)$, must undo this.

Let us check

It would subtract 5 and then divide by 3.

$$f^{-1}(x) = (x-5)/3$$

Now let us check it with the output 11:

$$f^{-1}(11) = (11-5)/3 = 2 \text{ which is our original input.}$$

Now let us see the formal definition of inverse function.

Definition: Let $f: A \rightarrow B$ be a bijective mapping. Then the mapping $g: B \rightarrow A$ which associates to each element $b \in B$ the unique element $a \in A$, such that $f(a) = b$, is called the inverse mapping of the mapping $f: A \rightarrow B$.

Remember, by definition a function to have an inverse, it must be **bijective**, meaning it is both **one-to-one** (each output comes from only one input) and **onto** (it covers all possible outputs).

Hence, if $f: A \rightarrow B$ is a one-one onto mapping, then $f^{-1}: B \rightarrow A$ exists, and is also 1-to-1.

For example, $A = \mathbb{R} \setminus \{3\}$ and $B = \mathbb{R} \setminus \{1\}$, and the function $f: A \rightarrow B$ is defined by $f(x) = (x-2)/(x-3)$.

We can see that f is a one-to-one and onto function.

$\therefore f$ inverse exists.

To find $f^{-1}(x)$ the following steps are required:

1. Replace $f(x)$ by y in the equation describing the function.

Here you will get $y = (x-2)/(x-3)$.

2. Interchange x and y . In other words, replace every x by y , and vice versa.

Here you will get $x = (y-2)/(y-3)$.

3. Solve for y .

- 4 Replace y by $f^{-1}(x)$

By applying these steps, we get $f^{-1}(x) = (3x-2)/(x-1)$.

Example:

Let's find the inverse of the function $f(x)=3x+5$ (which we have used above for verification the concept)

Step 1: Replace $f(x)$ with y .

$$y = 3x+5$$

Step 2: Swap x and y .

$$x = 3y+5$$

Step 3: Solve for y . Subtract 5 from both sides:

$$x-5 = 3y$$

Divide both sides by 3:

$$(x-5)/3 = (3y)/3$$

$$y = (x-5)/3$$

Step 4: Replace y with $f^{-1}(x)$.

$$f^{-1}(x) = (x-5)/3$$

The inverse of $f(x) = 3x+ 5$ is $f^{-1}(x) = (x-5)/3$.

The concept of inverse function is used in cryptography & security (encryption and decryption), solving mathematical equations, undo functionality in general computer applications (command (Ctrl+Z) in software is an application of an inverse function), computer graphics etc.

Check Your Progress -3

1. Which of the following kinds of function would you use to provide photo identity numbers and why?

i) Constant function, ii) one-to-one function, and iii) identity function.

.....

2. Explain why $f: Z \rightarrow Z: f(x) = x^2$ is onto? Domain and range of f is Z .

.....

3. Explain whether function defined by $f(x) = x^2 - 2x + 1$ is onto function or not?

.....

4. Find f^{-1} of rule $f : f(x) = x^3 - 3$.

.....

5. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 1$ then find $f^{-1}(17)$

.....

3.7 SUMMARY

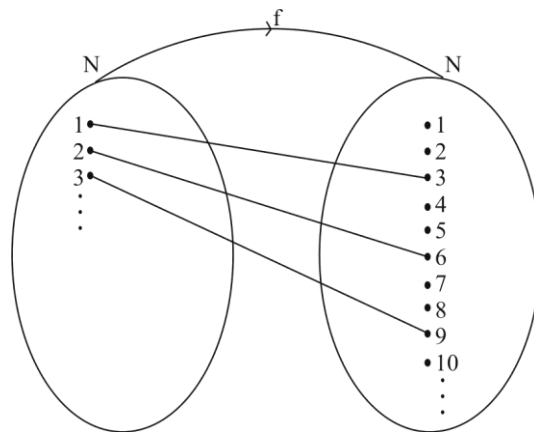
This unit provides a foundation of Functions, beginning with basic definitions like quantities and intervals before moving into the mechanics of how sets interact. This unit explains various ways to represent functions—whether through mapping diagrams, pictorial graphs, or algebraic operations and also introduces monotonicity, which describes whether a function consistently increases or decreases. Further, this unit focuses on the classification and reversibility of functions based on their mapping properties. It distinguishes between injective (one-to-one), surjective (onto), and bijective functions, using both algebraic definitions and geometric interpretations. Finally, the unit concludes by explaining how to find the inverse of a function.

3.8 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Answer to Check Your Progress- 1

1.
 $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by
 $f(x) = 3n, \quad n \in \mathbb{N}$
 $\therefore f(1) = 3, f(2) = 6, f(3) = 9$, and so on. See Fig. 3.6

Functions



Domain of the function $f = \{1, 2, 3, \dots\} = N$

Range of the function $f =$ Set of only those values for which function is defined $= \{3, 6, 9, \dots\}$

Co-domain $=$ Set of all values of $Y = \{1, 2, 3, \dots\} = N$

2. $f: R \rightarrow R$ is defined by

$$f(x) = 3^x, \quad x \in R$$

(ii) Since $f(x)$ is defined for all $x \in R$

$\therefore$ domain of $f = R =$ set of all real numbers

(iii) We know that

$$3^x > 0 \quad \forall x \in R$$

$$\text{i.e. } f(x) > 0 \quad \forall x \in R$$

$$\therefore \text{range of } f = (0, \infty)$$

3.

(i) Domain $= R$, Range $= \{x \in R \text{ and } x \leq 0\}$.

(ii) Domain $= [0, 3]$, Range $= [0, 3]$.

4. i) $f: N \rightarrow N$ is not a function, as $f(0) = -3 \notin N$, the codomain of f .

ii) $f: N \rightarrow Z$, is a function.

5. Given function is

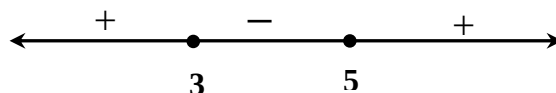
$$f(x) = \sqrt{(x-3)(5-x)}, \quad x \in R$$

For $f(x)$ to be real, the quantity under the square root should be non negative and hence

$$(x-3)(5-x) \geq 0$$

$$\Rightarrow -(x-3)(x-5) \geq 0$$

$$\Rightarrow (x-3)(x-5) \leq 0$$



Now, when we take x less than 3, the L.H.S. comes out to be $(-ve)(-ve) = +ve$, hence does not satisfy the inequality.

Also, if we take x greater than 5, the L.H.S. comes out to be $(+ve)(+ve) = +ve$, and hence this value does not satisfy the inequality.

But, if we take $3 < x < 5$, the L.H.S. becomes $(+ve)(-ve) = -ve$ and hence satisfies the inequality.

$$\Rightarrow x \in [3, 5]$$

$\therefore$ domain of $f = [3, 5]$

Also,

$$f(3) = \sqrt{(3-3)(5-3)} = \sqrt{0 \times 2} = \sqrt{0} = 0$$

$$f(4) = \sqrt{(4-3)(5-4)} = \sqrt{1 \times 1} = 1$$

$$f(5) = \sqrt{(5-3)(5-5)} = \sqrt{2 \times 0} = \sqrt{0} = 0$$

6. Here $f(x)$ takes real values for all $x \in \mathbb{R}$, except at $x = -3$

$\therefore$ domain of $f = \mathbb{R} - \{-3\}$

and $f(x)$ cannot take zero values at any real number,

$\therefore$ range of $f = \mathbb{R} - \{0\}$

$$\text{Also, } f(0) = \frac{1}{0+3} = \frac{1}{3}, f(2) = \frac{1}{2+3} = \frac{1}{5}$$

$f(-3)$ is not defined because $x = -3$ is not a point of the domain of f .

7. Not every relation is a function. For example, this relation does not satisfy the property that:

Each element of A must have been assigned to one element in B .

If $a \in A$ is assigned $b \in B$ and $a \in A$ is assigned $b' \in B$, then $b = b'$

That is why relations that don't satisfy the above properties are not a function

8. We can see that the code has no effect on the value of $n < 0$. In the While loop, the value of n is halved whenever it is even. If n becomes odd before reaching 1, the second part of the while loop is invoked, and n remains odd and increases forever.

This shows that $f: \mathbb{N} \rightarrow \mathbb{N}$ is the function defined by :

$f(n) = n/2$ if n is even and $2n+1$ if n is odd

Note : for $f(1)$ the loop will not be executive, infact $n > 1$ is the condition for loop execution.

Answer to Check Your Progress- 2

1.

i) $(f+g)(x) = 1/x + x^3 + 2$

ii) $(f-g)(x) = 1/x - (x^3 + 2)$

iii) $(f.g)(x) = (1/x)(x^3 + 2)$

iv) $(f/g)(x) = 1/(x(x^3+2)) \forall x \in \mathbb{R}$.

2.

i) $(f+g)(x) = \sqrt{x+1} + x^3 \forall x \geq -1$

ii) $(f-g)(x) = \sqrt{x+1} - x^3 \forall x \geq -1$

iii) $(f.g)(x) = (\sqrt{x+1}). x^3 \forall x \geq -1$

iv) $(f/g)(x) = \sqrt{x+1} / x^3 \forall x \geq -1, x \neq 0$

v) $(f \circ g)(x) = f(x^3) = \sqrt{x^3+1} \forall x \geq -1$

3. Let us find the composite functions $(f \circ g \circ h)(x)$ and then $(g \circ f)(x)$.

Given:

$$f(x) = x^2$$

$$g(x) = x + 2$$

$$h(x) = x^3 + x + 2$$

A. Calculating $(g \circ f)(x)$:

The composition $(g \circ f)(x)$ means we apply function f first, and then apply function g to the result.

We will start with the outer function: $g(x) = x + 2$

1. Substitute $f(x)$ into $g(x)$ in place of x : $g(f(x)) = f(x) + 2$

2. Replace $f(x)$ with its definition, x^2 : $g(f(x)) = x^2 + 2$

3. So, the final result is: $(g \circ f)(x) = x^2 + 2$

B. Calculation of $(f \circ g \circ h)(x)$

This composition means we work from the inside out: first $h(x)$, then $g(x)$, and finally $f(x)$. It's calculated as $f(g(h(x)))$.

i) First, find the inner composition, $g(h(x))$: $g(h(x)) = h(x) + 2$

Substitute the definition of $h(x)$: $g(h(x)) = (x^3 + x + 2) + 2$

$$g(h(x)) = x^3 + x + 4$$

ii) Now, substitute this result into the outer function, $f(x)$:

$$f(g(h(x))) = (g(h(x)))^2$$

Replace $g(h(x))$ with the expression we just found:

$$f(g(h(x))) = (x^3 + x + 4)^2$$

So, the composite function is: $(f \circ g \circ h)(x) = (x^3 + x + 4)^2$

Answer to Check Your Progress- 3

1. One-to-one function will be used for providing identity card number, because each person must have unique identity numbers.
2. Function $f(x) = x^2$ is one-to-one because for every value of x , x^2 will be a number that is different for different x . Hence, $f(x) = x^2$ is one-one mapping.
3. The function $f(x) = x^2 - 2x + 1$ is **not** an onto function. Because a function is **onto** (or surjective) if its **range** is equal to its **codomain**. For this function, the codomain is typically assumed to be all real numbers, $\mathbf{R}$. However, the range of this function is restricted.
This function can be simplified by recognising that it is a perfect square:
$$f(x) = x^2 - 2x + 1 = (x - 1)^2$$

Since the square of any real number $\mathbf{R}$ is always non-negative, the output of this function can never be a negative number. The minimum value of $f(x)$ is 0, which occurs when $x = 1$.
Therefore, the **range** of the function is $[0, \infty)$, which is not equal to the codomain of $\mathbf{R}$. For instance, there is no real number x for which $f(x) = -9$

4. Step 1: $y = x^3 - 3$

$$\text{Step 2: } x = y^3 - 3$$

$$\text{Step 3: } y = \sqrt[3]{x + 3}$$

$$\text{Step 4: } f^{-1}(x) = \sqrt[3]{x + 3}$$

5. To find the value of $f^{-1}(17)$, we need to determine which input value, x , results in an output of 17 for the function $f(x)=x^2+1$.

In other words, we need to solve the equation: $f(x)=17$

Step-by-Step Calculation

1. Set up the equation: Substitute the function definition into the equation:
 $x^2+1 = 17$
2. Isolate the x^2 term: Subtract 1 from both sides of the

$$\text{equation: } x^2 = 17-1 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$$

Hence $x = 4$ or $x = -4$

The values of x for which $f(x)=17$ are 4 and -4. Therefore, $f^{-1}(17)=\{4,-4\}$.

It's important to note that since two different input values (4 and -4) produce the same output (17), the function $f(x)=x^2+1$ is not a one-to-one function over the real numbers, and therefore **does not have a true inverse function** unless its domain is restricted. However, the set of values that map to 17 is $\{4,-4\}$.

3.9 REFERENCES AND FURTHER READINGS

1. Discrete Mathematical Structures By Bernard Kolman , Robert C. Busby , Sharon Cutler Ross , PHI 2002
2. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
3. Mathematics Class XI By Ravi Prakash , GRB Books, Edition 2025
4. MCS-013: Discrete Mathematics
Unit-1 Sets, Relations and Functions
(<https://egyankosh.ac.in/handle/123456789/1140>)
5. MEC-203 : Quantitative Methods
Unit-3 Relations and Functions
(<https://egyankosh.ac.in/handle/123456789/98141>)
6. MST-001: Foundation in Mathematics and Statistics
Unit-2 Functions
(<https://egyankosh.ac.in/handle/123456789/92122>)